

Effect of Prompt Length on AI Response Quality: An Evidential Reasoning Approach

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Abstract—Prompt length is one of the most direct levers a user has over the behaviour of a large language model (LLM), yet its effect on response quality is reported inconsistently. Most evaluations treat response quality as a single deterministic value rather than a multidimensional, uncertain judgment. This pilot study examines how increasingly detailed and structured prompts (short, medium, and long) affect perceived AI response quality across five criteria: perceived accuracy, relevance, completeness, clarity, and reasoning quality. We use a within-subjects design in which 114 respondents (predominantly undergraduate students) rated AI responses to a single FIFA World Cup prediction task in two data-collection phases (40 respondents rated answers from their own AI tools; 74 rated one fixed answer set). An Evidential Reasoning (ER) approach, applied to respondent-level ratings with judgement-based illustrative criterion weights, aggregates uncertain multi-criteria judgments into belief distributions with an explicit ignorance mass. Friedman tests with Bonferroni-corrected Wilcoxon signed-rank comparisons show that the long prompt received significantly higher ratings than the short prompt for accuracy, completeness, clarity, and reasoning quality, and higher ratings than the medium prompt for accuracy, clarity, and reasoning quality, with no differences for relevance. Omnibus effect sizes were small (Kendall's W ranged from 0.05 to 0.08); pairwise Z-based effect sizes for significant comparisons ranged from 0.25 to 0.39. Under ER aggregation, the belief in the top two grades rises from 42.5 percent (Short) to 46.5 percent (Medium) and 58.1 percent (Long), and the composite ordering from Short to Medium to Long is preserved under alternative criterion weights and under two assumed rater-reliability mappings. Because the long prompt also adds specificity and explicit reasoning instructions, and the first 40 respondents rated answers from their own AI tools, the effects cannot be attributed to length alone; in the fixed-answer subsample of 74 respondents, only the Short-versus-Long difference remained significant. Multi-task replication with diverse samples and expert-elicited weights is needed before broader conclusions can be drawn.

Index Terms—Prompt Engineering, Large Language Models, Prompt Length, Pilot Study, Evidential Reasoning, Response Quality Evaluation, Uncertainty Modelling

STATEMENT ON THE USE OF AI ASSISTANCE

Generative AI was used in two ways. ChatGPT (OpenAI) was used to draft the three prompts and to generate the three fixed AI responses shown to respondents in the second data-collection phase (Section III-C). Claude (Anthropic) assisted with Python code for the statistical and ER computations, with the web-based research interface, with identifying candidate references (independently verified by the authors), and with language editing. All results were computed from the raw respondent-level data and can be reproduced from the released

data and code. The authors reviewed and take full responsibility for all content.

I. INTRODUCTION

Large Language Models (LLMs) such as GPT, Gemini, Claude, and LLaMA have become central to how people search for information, write, code, and make decisions. Because the model itself is typically not directly modifiable by an end user, the prompt is the single most accessible lever for shaping the quality of the output.

Among the many properties of a prompt that can be varied—wording, structure, specificity, the presence of examples—prompt length is one of the simplest to manipulate and one of the most frequently discussed in both academic literature and everyday practice. A short prompt is fast to write but may leave the model without sufficient context; a long prompt provides more grounding but risks introducing irrelevant information or competing instructions.

The empirical evidence on this trade-off is mixed. Domain-specific studies report that long, context-rich prompts generally outperform short ones [1], while research on long-context reasoning shows that simply having a large context window does not guarantee effective reasoning [2], [3]. A second gap concerns how AI response quality is measured: the overwhelming majority of existing studies rely on automated, single-dimensional metrics that say little about whether a response is clear, well-reasoned, complete, or genuinely useful.

The present paper addresses both gaps. It reports a single-task pilot study in which 114 participants (predominantly undergraduates) rated AI responses to a FIFA World Cup prediction question posed with three increasingly detailed and structured prompts (short, medium, long), evaluated across five quality criteria. Because the prompts differ in detail and structure as well as length, we treat the manipulated factor as increasingly detailed and structured prompting, with “Short”, “Medium”, and “Long” labelling its three levels. We additionally apply an Evidential Reasoning approach, with judgement-based illustrative criterion weights, to aggregate multi-criteria, multi-rater judgments into belief distributions. All inferential claims are grounded in formal statistical tests (Friedman and Wilcoxon), verified against the raw respondent-level data, and the study's scope is explicitly bounded: these findings should not be generalised beyond this task type and sample without replication.

The remainder of this paper is organised as follows. Section II reviews related work. Section III describes the methodology. Section IV presents results. Section V presents the ER evaluation approach and its numerical illustration. Section VI discusses implications and limitations. Section VII concludes.

II. LITERATURE REVIEW

A. Multi-Criteria Decision Making and Prompt Engineering

The Analytic Network Process (ANP), introduced by Saaty as a generalisation of the Analytic Hierarchy Process, is a multi-criteria decision-making method designed for problems in which evaluation criteria are not independent [4]. ANP represents a decision problem as a network of clusters and elements, allowing both inner and outer dependencies. For AI-generated text evaluation, ANP is attractive because response-quality dimensions—accuracy, relevance, clarity, completeness, and reasoning quality—plausibly influence one another rather than acting independently (an assumption of this pilot study rather than an established finding). The literature integrating ANP with LLM evaluation remains sparse; in this pilot study the criterion weights used for the composite score are judgement-based and analytically constructed (Section III-D), not elicited from survey respondents and not derived through a formal ANP procedure.

LLMs are Transformer-based models trained primarily through next-token prediction, and among the factors shaping output quality, the prompt is the only one a typical end user can control directly. Prompt engineering is the practice of designing prompts to improve accuracy, relevance, clarity, and usefulness while reducing hallucination [5]–[7], via techniques such as Chain-of-Thought (CoT) prompting [8] and Few-Shot/Zero-Shot approaches [9], [10]. Most studies evaluate success through task performance metrics alone, rarely applying structured, multi-criteria evaluation frameworks. Evaluation research is, however, moving toward multi-metric assessment: HELM reports seven metrics (accuracy, calibration, robustness, fairness, bias, toxicity, and efficiency) across core scenarios [11], recent surveys organise LLM evaluation by what, where, and how to evaluate [12], and LLM-as-a-judge studies compare model answers with human preferences [13].

B. Effect of Prompt Length and Long-Context Limitations

Liu, Wang, and Willard [1] examined prompt length effects across nine domain-specific tasks, finding that Long Instructions improved F1-scores on all nine tasks while Short Instructions reduced them on all nine. Sabbatella *et al.* [14] applied Bayesian optimisation over a continuous relaxation of the discrete prompt space, noting that the search space grows with prompt-sequence length, and in preliminary RoBERTa experiments on six benchmarks analysed the trade-off between search-space size, accuracy, and wall-clock time.

Levy, Jacoby, and Goldberg [2] introduced the FLenQA benchmark, isolating input-length effects by embedding relevant content within irrelevant padding text. Across several LLMs, reasoning accuracy fell from approximately 92% in short contexts to approximately 68% in longer contexts, with degradation beginning around 3,000 tokens—far short of advertised context windows. The “lost in the middle” effect [3] further shows that the position of relevant information within a long context substantially affects performance.

C. Research Gap

Across the reviewed literature, three gaps recur: (1) AI response quality is treated as one-dimensional; (2) interdependence among quality dimensions is not modelled; (3) uncertainty and disagreement in human judgment are largely ignored. To our knowledge, no

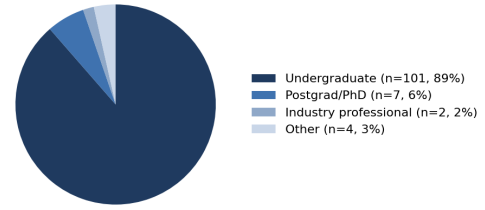


Fig. 1. Participant composition by role ($N = 114$).

unified framework combines importance-weighted criteria, ER-based uncertainty aggregation, and formal statistical testing of prompt-length effects. This pilot study begins to address all three gaps at small scale and identifies the requirements for a full-scale replication.

III. METHODOLOGY

A. Research Design

This study uses a mixed-methods within-subjects design combining (a) a cross-sectional survey of AI tool users’ general attitudes toward prompt length, and (b) a controlled within-subjects rating exercise in which the same underlying question was posed to an AI assistant at three prompt levels (short, medium, and long), rated by every participant on five quality criteria.

B. Participants

A total of 114 participants completed the survey. The majority were undergraduate students ($n = 101$; 89%), with smaller numbers of postgraduate/PhD students ($n = 7$), industry professionals ($n = 2$), and others ($n = 4$). Regarding AI tool experience, 50 participants reported 1–2 years of use, 45 reported more than 2 years, 13 reported 6 months to 1 year, and 6 reported less than 6 months. Only two background variables were collected: primary role (Fig. 1) and duration of AI tool use. No other demographic information (e.g., age, gender, or department) was recorded.

Scope note: The sample is predominantly undergraduate and was recruited from a single institution. All inferential results should be interpreted as pertaining to this sample and task type only; generalisation requires independent replication. The general-attitude questions (Section IV-A) were part of the first version of the form and were answered by the first 40 respondents only; all rating items and the remaining background items have $N = 114$.

C. Three-Tier Prompting Task

To examine how prompts of different length and detail affect responses while holding the underlying task constant, each participant was shown three AI-generated responses to the same general question (e.g., “Who is more likely to win the next FIFA World Cup?”), each elicited using a prompt of different length:

- **Short prompt:** a single, minimal-context question with no supporting instructions.
- **Medium prompt:** the same question with a brief request to explain the answer with reference to team strength, performance, and players.
- **Long prompt:** the same question expanded into a multi-part instruction naming six candidate teams and asking for a detailed, multi-factor analysis covering performance, player quality, recent results, teamwork, and tactical strength, followed by a reasoned final prediction.

Response generation. For the fixed answer set used in Phase 2, the three prompts were drafted with ChatGPT and the three responses were generated with GPT-5.6 Luna (version 5.6) through the free-tier

ChatGPT web interface on 17 June 2026. Custom instructions and memory were not enabled. Sampling settings such as temperature are not exposed by this interface, so default settings were used. The three prompts were submitted in the same chat session, in the order Short, Medium, Long, with one response generated per prompt and used without editing. The prompts contained 11, 24, and 49 words (approximately 12, 28, and 62 tokens; counted with the tiktoken o200k_base tokenizer, because the ChatGPT interface does not report token counts and the exact tokenizer of GPT-5.6 Luna is not public). Response length also increased with prompt detail (32, 85, and 126 words), so response length is a further factor that cannot be separated from prompt length.

Data-collection phases. The form was administered in two phases. In Phase 1 (10–18 June 2026; $n = 40$), each respondent entered the prompts into an AI tool of their own choice, pasted the answer obtained, and rated it; the answers therefore differed across respondents, and the tools and models used were not recorded. After the form was updated on 19 June, Phase 2 (from 22 June 2026; $n = 74$) displayed the same three fixed answers described above to every respondent, together with the prompt text and condition label. The general-attitude questions were asked in Phase 1 only. Ratings are pooled across phases in the main analysis; Section IV-C reports the two phases separately.

For each response, participants rated five criteria on a 1–5 scale: perceived Accuracy, Relevance, Completeness, Clarity, and Reasoning Quality. Accuracy is a perceived rating throughout this paper. In both phases, the three conditions were presented in the same fixed order (Short, Medium, Long), each labelled with its condition name and prompt text. This single-task, single-topic design is appropriate for a pilot study establishing whether the ER approach is feasible, but is not sufficient to generalise findings across task types (Section VI).

D. Criterion Weighting: Source and Rationale

The criterion weights used for the composite score are judgement-based and analytically constructed; they were not elicited from the 114 respondents. The weights (Accuracy 0.28, Relevance 0.22, Clarity 0.20, Reasoning Quality 0.18, Completeness 0.12) reflect the frequency with which respondents selected each criterion as most important (Section IV-A), adjusted by the authors’ judgment of the relative importance of the criteria. They should be read as illustrative assumptions, not as empirically derived or expert-elicited values. The weights are not a direct transcription of the selection counts: for example, Reasoning Quality is weighted above Completeness (0.18 versus 0.12) although Completeness was selected more often (23 selections). They are not the output of a formal ANP pairwise-comparison, consistency, and supermatrix procedure; such an elicitation is left for future work (Section VII). A sensitivity analysis (Section V-D) shows that the composite ordering is robust to the weighting scheme.

One respondent did not rate Reasoning Quality under the Medium-prompt condition; this respondent is included in all single-criterion analyses where their data is complete ($N = 114$ for all criteria except Reasoning Quality, $N = 113$) and excluded only from composite-score calculations ($N = 113$).

E. Statistical Analysis Plan

Given the within-subjects design and ordinal data, non-parametric tests were used throughout: Friedman tests for each criterion and the weighted composite; Wilcoxon signed-rank post-hoc pairwise comparisons (Short vs. Medium, Short vs. Long, Medium vs. Long) with Bonferroni correction (adjusted $\alpha = .05/3 = .0167$) where the

TABLE I
MEAN (SD) RATINGS BY CRITERION AND PROMPT-LENGTH CONDITION,
WITH FRIEDMAN TEST RESULTS

Criterion	Short	Medium	Long	χ^2	p
Perceived accuracy	3.32 (0.88)	3.29 (0.99)	3.51 (1.00)	12.173	.0023
Relevance	3.42 (1.01)	3.42 (1.02)	3.42 (1.09)	0.008	.9959
Completeness	3.27 (0.98)	3.37 (1.02)	3.55 (0.96)	12.354	.0021
Clarity	3.27 (0.95)	3.42 (0.98)	3.64 (0.96)	18.034	<.0001
Reasoning	3.21 (0.95)	3.28 (0.96)	3.52 (1.07)	15.358	.0005
Weighted composite	3.32	3.37	3.54	11.603	.0030

Note: The composite uses the judgement-based weights (Section III-D) and complete-case data ($N = 113$). Kendall’s W : Accuracy .053, Completeness .054, Clarity .079, Reasoning .068, Composite .051 (all small effects).

Friedman test was significant, so that the correction family is the three pairwise comparisons within each criterion; effect sizes as Kendall’s W (Friedman) and the Z -based effect size $r = |Z|/\sqrt{n}$ (Wilcoxon); a sensitivity analysis comparing weighted and equal-weighted composites; percentile bootstrap 95% confidence intervals (5,000 resamples) for composite differences; and the ER aggregation of Section V. All analyses were conducted in Python 3 (pandas, scipy.stats) and re-executed against the raw data to confirm reproducibility.

IV. RESULTS

A. General Attitudes Toward Prompt Length

Among the 40 respondents who answered the attitude questions ($n = 40$), a clear majority believed that prompt length affects response quality (18 “Yes, strongly,” 13 “Yes, somewhat,” 5 neutral or unsure, 4 “No”). Medium-length prompts (3–5 sentences) were the most common preferred length (17 of 40), while 15 respondents felt the ideal length “depends on the task.” Mean satisfaction was higher for long/detailed prompts ($M = 3.6$) than for short prompts ($M = 3.2$) ($n = 40$). Across all 114 respondents, Accuracy was selected most often as the most important quality (69 selections), followed by Relevance (45), Clarity (42), Completeness (23), and other factors (12) (multiple selections were allowed, so the counts sum to more than 114); this ordering informed the criterion weighting in Section III-D. Most respondents acknowledged that their judgment varies with mood or context at least “somewhat” (64) or “a lot” (20) ($N = 114$)—evaluator-level uncertainty that point-estimate averaging cannot represent and that the ER analysis in Section V explicitly models.

B. Descriptive and Inferential Statistics

Table I reports mean ratings, standard deviations, and Friedman test results ($N = 114$ except Reasoning Quality and the weighted composite, $N = 113$) by prompt-length condition. The weighted composite rises monotonically from Short (3.32) to Medium (3.37) to Long (3.54). The improvement is not uniform: Clarity shows the largest gain ($\Delta = 0.37$), followed by Reasoning Quality ($\Delta = 0.31$) and Completeness ($\Delta = 0.28$), while Relevance remains flat ($p = .996$, not significant)—indicating no detectable prompt-length effect on topical relevance. All other criteria and the composite show statistically significant differences across conditions (all $p \leq .004$). Fig. 2 plots the condition means with 95% bootstrap confidence intervals.

For each significant Friedman result, pairwise Wilcoxon signed-rank tests were conducted with Bonferroni correction (Table II). The clearest and most consistent effect is a Short-to-Long improvement (all $p \leq .005$); no Short-vs.-Medium comparison reached corrected significance for any criterion. Medium-to-Long differences reached significance for Accuracy ($p = .007$), Clarity ($p = .009$), and Reasoning Quality ($p < .001$), but not Completeness after correction. Relevance showed no significant difference across any pair. Pairwise effect sizes ($r = |Z|/\sqrt{n}$) ranged from $r = 0.25$ (Clarity, Medium vs. Long) to $r =$

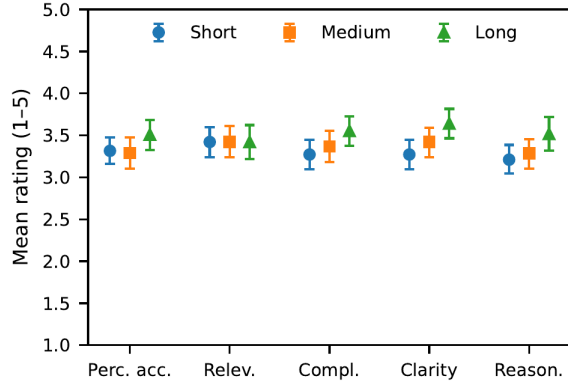


Fig. 2. Mean ratings with 95% bootstrap confidence intervals (5,000 resamples) by criterion and prompt condition ($N = 114$; $N = 113$ for Reasoning, Medium). Intervals describe each mean separately; the within-subject comparisons are tested in Table II. Relevance remains flat while Clarity, Reasoning, Completeness, and Perceived accuracy rise with prompt detail.

TABLE II
WILCOXON SIGNED-RANK POST-HOC COMPARISONS
(BONFERRONI-CORRECTED)

Criterion	Comp.	W	p	r
Accuracy	S vs M	597.0	.8677	0.016
	S vs L	441.0	.0049**	0.264
	M vs L	307.0	.0066**	0.254
Relevance	S vs M	733.0	.9315	0.008
	S vs L	903.0	.9237	0.009
	M vs L	656.0	.9446	0.007
Completeness	S vs M	610.0	.2206	0.115
	S vs L	668.0	.0028**	0.280
	M vs L	331.5	.0240 [†]	0.211
Clarity	S vs M	454.0	.0525	0.182
	S vs L	378.0	<.0001**	0.387
	M vs L	421.5	.0087**	0.246
Reasoning	S vs M	539.5	.4404	0.073
	S vs L	427.0	.0013**	0.301
	M vs L	218.5	.0004**	0.330

** $p < .0167$ (significant after Bonferroni correction). [†]Reached uncorrected significance ($p = .024$) but not corrected significance. $r = |Z|/\sqrt{n}$ (Z -based effect size). S = Short, M = Medium, L = Long.

0.39 (Clarity, Short vs. Long)—small-to-moderate, consistent with the small Kendall’s W values at the omnibus level.

C. Robustness Across Data-Collection Phases

Because the first 40 respondents rated answers from their own AI tools whereas the remaining 74 rated one fixed answer set (Section III-C), the weighted composite was re-analysed separately for each phase (complete cases). In Phase 2 ($n = 74$; means 3.24, 3.33, and 3.41 for Short, Medium, and Long), the Friedman test was significant ($\chi^2 = 6.374$, $p = .041$) and Short vs. Long remained significant after Bonferroni correction (adjusted $p = .011$), whereas Short vs. Medium (adjusted $p = .22$) and Medium vs. Long (adjusted $p = .40$) did not. In Phase 1 ($n = 39$; means 3.48, 3.44, and 3.78), Short vs. Long (adjusted $p = .009$) and Medium vs. Long (adjusted $p = .006$) were both significant. The Long-minus-Short composite difference did not differ detectably between phases (0.30 vs. 0.17; Mann-Whitney $p = .49$). The Short-to-Long improvement is therefore present in both phases, whereas the pooled Medium-to-Long difference was not significant in the fixed-answer subsample and should be interpreted cautiously.

TABLE III
EVALUATION HIERARCHY: MAIN CRITERIA AND REPRESENTATIVE
SUB-CRITERIA

Main criterion	Representative sub-criteria
Perceived accuracy	Perceived factual correctness; consistency with known information
Relevance	On-topic focus; direct address of the question asked and identification of the relevant points
Completeness	Coverage of the relevant factors and completion of the ideas introduced
Clarity	Clear, well-articulated expression that is easy to follow
Reasoning Quality	Logical flow; structured, coherent explanation and overall analytical flow

V. EVIDENTIAL REASONING EVALUATION

A. Evaluation Hierarchy

Before assigning criterion weights, it is useful to make the evaluation hierarchy explicit. Table III summarises the goal, the five main quality criteria (to which the weights are applied), and representative sub-criteria describing what each main criterion means in measurable terms. This hierarchy underlies the approach: criterion weights are applied to the five main criteria (Section V-B); rater judgments are represented as belief distributions over five grades; and the ER algorithm combines these distributions across raters and criteria into an overall belief distribution per condition (Section V-C).

B. Rationale and Approach

The descriptive means and inferential tests in Section IV establish the direction and significance of prompt-length effects, but mask two sources of uncertainty: (a) individual raters disagree, and a simple mean discards information about how much agreement underlies it; and (b) the five criteria are not equally important. A defensible approach should weight criteria by relative importance and represent each rater’s judgment as a distribution of belief rather than a single collapsed number.

Inspired by the ANP idea that criteria should be weighted by relative importance rather than treated as equal [4], the five criteria receive the normalised weights $w = (0.28, 0.22, 0.12, 0.20, 0.18)$ for Accuracy, Relevance, Completeness, Clarity, and Reasoning Quality (judgement-based, not respondent-elicited). ER has previously been applied to multi-attribute assessment under uncertainty, for example in the evaluation of job offers [15].

Let $H_1, \dots, H_5$ denote the grades corresponding to ratings 1–5 and H the whole frame (ignorance). Rater k ’s rating $x_{k,i}$ on criterion i is a basic probability assignment (BPA)

$$\begin{aligned} m_{k,i}(H_n) &= r_k \cdot \mathbf{1}[x_{k,i} = n], \\ m_{k,i}(H) &= 1 - r_k, \end{aligned} \quad (1)$$

where $r_k \in [0, 1]$ is the rater’s reliability; a missing rating has $m_{k,i}(H) = 1$. Rater-level beliefs are pooled per criterion, $\beta_{n,i} = \frac{1}{N} \sum_k m_{k,i}(H_n)$ and $\beta_{H,i} = \frac{1}{N} \sum_k m_{k,i}(H)$, because directly multiplying 114 conflicting BPAs with Dempster’s rule would produce spuriously confident results. The five pooled criterion-level distributions are then combined with the recursive ER algorithm of Yang and Xu [16], which is built on Dempster–Shafer evidence combination [17], using the weights w_i . The result is an overall belief distribution $\{\beta_n\}_{n=1}^5$ with ignorance β_H for each prompt-length condition. With utility $u(H_n) = n$, the expected utility lies in the interval

$$\left[\sum_n \beta_n n + \beta_H \cdot 1, \sum_n \beta_n n + \beta_H \cdot 5 \right]. \quad (2)$$

In the baseline analysis all raters are treated as fully reliable ($r_k = 1$), so ignorance arises only from the single missing rating.

TABLE IV
ER-AGGREGATED BELIEF DISTRIBUTION PER PROMPT-LENGTH
CONDITION (BASELINE, $r_k = 1$)

Cond.	β_1	β_2	β_3	β_4	β_5	β_H	Utility
Short	.046	.092	.437	.351	.074	0	3.317
Medium	.052	.089	.392	.372	.093	.001	3.363–3.369
Long	.052	.065	.302	.447	.134	0	3.546

β_n : belief in grade H_n (rating n); β_H : ignorance. Weights as in Section V-B; $N = 114$ ($N = 113$ for Reasoning, Medium). Utility $u(H_n) = n$. The Medium utility is a range because the single missing Reasoning Quality rating contributes ignorance mass ($\beta_H = .001$).

TABLE V
SENSITIVITY ANALYSIS: JUDGEMENT-BASED VS. EQUAL WEIGHTING

Weighting scheme	Short	Medium	Long
Judgement-based weights	3.319	3.365	3.538
Equal weights	3.312	3.368	3.543

Note: weights = Acc .28, Rel .22, Cla .20, Rea .18, Com .12. Complete-case data, $N = 113$. Both schemes give the identical Short < Medium < Long ordering.

To represent evaluator-level uncertainty, we additionally set r_k from each rater’s self-reported judgment variability (Never = 1.00, Rarely = 0.95, Somewhat = 0.85, A lot = 0.70). These reliability values are illustrative assumptions, not estimates from the data.

C. Numerical ER Illustration

Table IV reports the ER-aggregated belief distribution per condition under the baseline analysis. The belief in grades 4 or 5 rises from 42.5% (Short) to 46.5% (Medium) and 58.1% (Long)—a pattern that a single mean does not show. The ER utility differs slightly from the plain weighted mean of Table I (e.g., 3.317 versus 3.306 for Short on all 114 raters) because ER is a non-linear aggregation. With the assumed reliability mapping, the ignorance mass is about 0.12 in every condition, and the utility intervals are [3.04, 3.54] (Short), [3.08, 3.58] (Medium), and [3.24, 3.74] (Long), with midpoints 3.29, 3.33, and 3.49. Under a more conservative mapping (1.00, 0.90, 0.75, 0.50), the ordering of midpoints is unchanged (3.27, 3.31, 3.45), although ignorance rises to about 0.21 and the intervals widen and overlap more. The ER results therefore support the direction of the effect but do not by themselves establish it: the main evidence remains the Friedman and Wilcoxon tests (Section IV), supported by bootstrap 95% confidence intervals for the composite differences (Long–Short [0.12, 0.31]; Long–Medium [0.08, 0.27]; Medium–Short [−0.03, 0.13]).

D. Sensitivity Analysis: Judgement-Based vs. Equal Weights

To assess whether the composite ordering depends on the weighting scheme, we compared the judgement-weighted composite against an equal-weighted composite (each criterion weighted 0.20), using complete-case data ($N = 113$; Table V). The Short < Medium < Long ordering is identical under both schemes, and composite scores differ by at most 0.007—the qualitative conclusion is robust to the choice of weighting scheme.

The ER analysis offers three things that mean comparison and Friedman tests alone cannot: (1) belief distributions that show how ratings are spread across grades rather than a single mean; (2) an explicit ignorance mass, so that missing or less reliable judgments widen the utility interval instead of being silently averaged away; and (3) a principled, transparent mechanism for combining evidence across criteria. Two limitations apply: the reliability mapping is assumed, and the weights are judgement-based rather than expert-elicited. The sensitivity analysis also shows that the weighting scheme has almost no practical influence on the composite (differences of at most 0.007), so the weights are not what drives the central result.

VI. DISCUSSION

The pattern observed—quality rising with prompt length, but unevenly across criteria, with Relevance flat and Clarity/Reasoning Quality showing the largest gains—is broadly consistent with [1], which found that longer prompts, supplying additional domain background, generally improved performance on domain-specific tasks, although those tasks and metrics differ from the perceived-quality ratings used here. The near-absence of a Short-to-Medium effect is notable: incremental lengthening produced little measurable benefit, whereas the step from medium to long—which added explicit multi-factor analytical structure—produced the clearest gains in Clarity and Reasoning Quality. This Medium-to-Long gain was not significant in the fixed-answer subsample (Section IV-C), so it should be treated as tentative.

The flat Relevance score ($p = .996$) is among the more practically important findings: additional prompt length neither helped nor harmed topical adherence, qualifying a simple “longer is better” narrative. A likely reason is that all three prompts ask the same question, so all responses stay on topic. Note that the long-context literature [2] concerns reasoning accuracy over inputs padded to thousands of tokens, whereas the prompts here contain only 12 to 62 tokens, so that literature does not directly bear on this result.

Effect sizes were small at the omnibus level (Kendall’s $W = 0.05$ – 0.08) and small-to-moderate for significant pairwise comparisons ($r = 0.25$ – 0.39); statistical significance should not be read as a large practical effect. The pilot is sufficient to detect a pairwise Short-vs.-Long difference in this sample, but effect-size estimates cannot be extrapolated to different task types, models, or populations.

Two interpretive cautions apply. First, the Long prompt differs from the Short and Medium prompts not only in length but also in specificity: it names six candidate teams and requests a multi-factor analysis and a reasoned final prediction. Higher ratings for Long may therefore reflect added structure and explicit reasoning instructions rather than length itself, and we refer to the manipulated factor as “increasingly detailed and structured prompting.” Second, “Accuracy” is a perceived rating: the target event (the next World Cup winner) cannot be verified when the ratings are given.

Limitations. The primary limitations are the single-task design (a speculative, sports-specific prediction task; generalisation to other domains, models, or prompt-length ranges is unknown) and the sample composition (89% undergraduate, single institution, not representative of domain experts or professional LLM users). Demographic information was limited to primary role and duration of AI tool use; age, gender, department, and other background variables were not collected, so differences across such groups could not be examined. Additionally, the criterion weights are judgement-based rather than elicited by a formal ANP procedure, the rater-reliability values used for the ignorance analysis are assumed, the prompt-length range is narrow (12 to 62 tokens, i.e., one sentence to one short paragraph), and the three prompts differ in content as well as length. The first 40 respondents rated answers produced by AI tools of their own choice, so the model, sampling settings, and content varied across them and were not recorded; only the remaining 74 respondents rated an identical answer set, and in that subset only the Short-vs.-Long difference remained significant (Section IV-C). The fixed Phase 2 responses come from a single model accessed through a consumer interface with non-configurable sampling settings and a single generation per prompt, so model- and sampling-specific variation is not captured. Because all three prompts were submitted in one chat session, earlier prompts and responses may have influenced later responses (e.g., the Long

response may build on context from the Short and Medium exchanges). Response length also grew with prompt detail (32, 85, and 126 words), which may itself have raised ratings of completeness and clarity. A related verbosity bias, a preference for longer answers, has been reported for LLM judges [13], and human raters may behave similarly. The exact prompts and responses are provided in the repository. All raters were presented with the conditions in the same fixed order (Short, Medium, Long) with the condition labels visible. Order and carry-over effects, and expectation bias (raters may have expected the more detailed prompt to yield a better response), therefore cannot be separated from the effect of prompt detail.

VII. CONCLUSION AND FUTURE WORK

This pilot study examined the effect of increasingly detailed and structured prompting (short, medium, and long prompts) on perceived AI response quality through a within-subjects design ($N = 114$, predominantly undergraduate students) and a single FIFA World Cup prediction task rated across five quality criteria. Friedman tests confirmed statistically significant differences across prompt-length conditions for perceived Accuracy, Completeness, Clarity, Reasoning Quality, and the weighted composite (all $p \leq .004$), while Relevance showed no significant effect ($p = .996$). Wilcoxon post-hoc tests identified the Long-vs.-Short comparison as the most consistent source of difference; no Short-vs.-Medium comparison reached significance for any criterion. The Short-to-Long improvement was present in both data-collection phases, whereas the Medium-to-Long difference was not significant in the fixed-answer subsample. A sensitivity analysis confirmed the composite ordering (Short < Medium < Long) is preserved under both weighting schemes, and ER aggregation showed the belief in the top two grades rising from 42.5% to 58.1%.

These results provide preliminary, within-sample evidence that increasingly detailed prompts matter for perceived response quality—specifically for depth-related criteria (Clarity, Reasoning, Completeness) rather than topical relevance—and illustrate how an Evidential Reasoning aggregation of respondent-level ratings, with an explicit ignorance mass, can complement formal inferential testing in LLM evaluation research. They should not be generalised beyond this single task and sample without replication.

Future work. Key next steps: (1) multi-task replication across four to six task types to estimate task $\times$ prompt-length interaction effects, with length varied independently of specificity; (2) expert-elicited criterion weights via a formal ANP pairwise-comparison instrument; (3) broader, more diverse samples including professional and expert users; (4) ER with empirically calibrated rater reliabilities on a larger sample; and (5) an extended prompt-length range approaching contemporary context windows, connecting these findings with the long-context degradation literature.

DATA AVAILABILITY

The survey instrument, raw respondent-level data, analysis code (including the ER script `er_analysis.py`), and the web-based research interface used in this study are openly available at <https://github.com/soroy-joyee-research-lab/prompt-length-ai-quality> and <https://response-link.vercel.app/>. The exact prompts and responses are in the `prompts/` folder of the repository.

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